SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

SIMATS ENGINEERING

SUB CODE & NAME: CSA17- ARTIFICIAL INTELLIGENCE

Exp 5: Write the python program for Missionaries Cannibal problem

CODE:

```
from collections import deque

# State: (missionaries_left, cannibals_left, boat_side)
# Boat side: 0 = left, 1 = right

def valid(m, c):
    if not (0 <= m <= 3 and 0 <= c <= 3):
        return False

    mr, cr = 3 - m, 3 - c

    return ((m == 0 or m >= c) and
            (mr == 0 or mr >= cr))

def solve():
    start = (3, 3, 0)
    goal = (0, 0, 1)

    queue = deque([(start, [start])])
    visited = {start}

    moves = [(1, 0), (2, 0), (0, 1),
```

```
            (0, 2), (1, 1)]

    while queue:
        state, path = queue.popleft()

        if state == goal:
            return path

        m, c, boat = state

        for dm, dc in moves:
            if boat == 0:
                new_state = (m - dm, c - dc, 1)
            else:
                new_state = (m + dm, c + dc, 0)

            if (new_state not in visited and
                    valid(new_state[0], new_state[1])):
                visited.add(new_state)
                queue.append((new_state, path + [new_state]))

    return None

path = solve()

if path:
    for state in path:
        print(state)
else:
```

print("No solution found")

OUTPUT:

